Evaluating the Impact of Individual Attitudes and Socio-Environmental Policies Across Welfare State Regimes in Addressing Climate Change

Diellza Veliu – Part 1: Hypothesis 1 & 2
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The paper examines levels of climate change awareness and the factors influencing people's decisions to adopt socio-environmental policies at both individual and state levels. It begins by analyzing individual concerns and participation in climate change discussions, then shifts to state-level analysis, observing the impact of socio-environmental policies on poverty reduction and renewable energy adoption. We used data from the 8th round (2016) of the European Social Survey (ESS), which surveyed 44,387 respondents across 23 European countries. The survey measured public attitudes toward climate change, energy preferences, and welfare state attitudes. The paper is organized around four hypotheses, for which we have selected the variables listed below.

Dependent variables:

Concern about Climate Change	**wrclmch:** Worried about climate change **clmthgt1:** Thought about climate change before **ccrdprs:** Personal responsibility to reduce climate change
Support for Environmental Policies	**inctxff:** Support for increasing taxes on fossil fuels. **sbsrnen:** Support for subsidizing renewable energy sources. **banhhap:** Support for banning the sale of energy-inefficient appliances. **eneffap:** Likelihood to buy energy efficient appliances
Poverty Prevention	**sbprvwdp:** Social Benefits Prevent Widespread Poverty

Independent variables:

Online Engagement	**netusoft:** Frequency of internet use. **netustm:** Amount of time spent online daily.
Socio-Economic Status – SES	**eisced -** Highest level of education **hinctnta -** Income level **pdwrk -** Paid work during last 7 days **rtrd -** Retired during last 7 days **uempla -** Unemployed last 7 days; actively job seeking

Control variables: Age; Gender; Country; Domicile; Design weight

Results Interpretation - Part 1: Diellza Veliu – 6612384

Hypothesis 1: *There are significant differences in the level of concern about climate change across individuals in different welfare state regimes.*

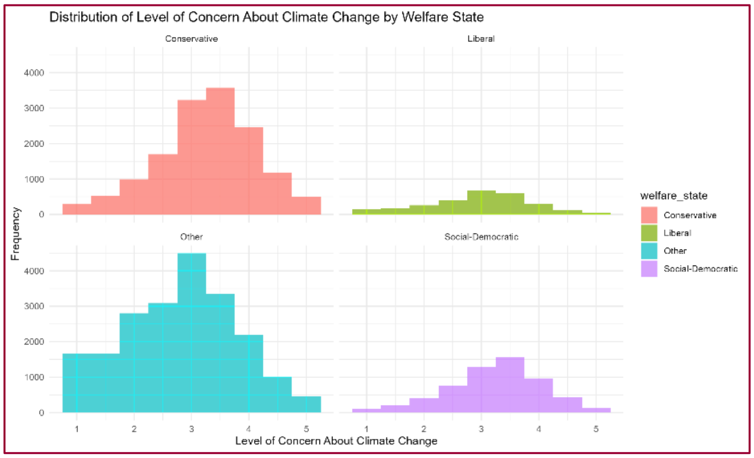
The table below reveals that individuals in Conservative and Social-Democratic welfare states exhibit the highest average concern about climate change, with mean concern levels of 3.27 and 3.23, respectively. This suggests that people in these regimes, which are typically characterized by strong social protection and proactive environmental policies, are more likely to worry about climate change. The median values for these regimes are both 3.50, further reinforcing that a significant portion of the population is indeed concerned. In contrast, individuals in Liberal welfare states show a slightly lower mean concern level of 2.95. I did not include the "Other" category in my analysis, as I focused on the three main welfare regimes based on Gøsta Esping-Andersen's model. The variability in concern levels is captured by the standard deviation (SD), where the Liberal regime (SD = 0.91) shows more dispersed concern levels, indicating a more heterogeneous population in terms of attitudes toward climate change.

Table: Descriptive Statistics of Climate Change Concern Across Welfare State Regime

welfare_state	*concern_mean*	*concern_median*	*concern_sd*	*concern_min*	*concern_max*
Conservative	3.27	3.50	0.86	1	5
Liberal	2.95	3.00	0.91	1	5
Other	2.83	3.00	0.98	1	5
Social-Democratic	3.23	3.50	0.83	1	5

The graph below reinforces the **descriptive statistics**, showing higher concern levels in Conservative and Social-Democratic welfare states, while Liberal regimes display more variability, with lower concern in Liberal states. This supports my first hypothesis of significant differences in concern across welfare state regimes.

Graph: Climate Change Concern Distribution by Welfare State Regimes



Additionally, I was keen on seeing if there is any difference in the level of concern about climate change between men and women. The cross-table analysis between gender and the level of concern about climate change reveals a noticeable pattern: Females generally show higher levels of concern about climate change compared to males, particularly at levels 3.5, 4, and 4.5. Males, on the other hand, are more evenly distributed across lower levels of concern. The chi-square test ($\chi 2$=114.048, p=0.000) confirms a significant association between gender and concern level, though the effect size is small (Cramer's V=0.036).

Crosstable between Gender and Level of Concern About Climate Change

gndr	*level_concern* 1	1.5	2	2.5	3	3.5	4	4.5	5	***Total***
Male	1087	1309	2190	2886	4465	4243	2683	1150	503	20516
	5.3 %	6.38 %	10.67 %	14.07 %	21.76 %	20.68 %	13.08 %	5.61 %	2.45 %	100 %
	53.57 %	52.19 %	50.45 %	49.21 %	46.95 %	46.97 %	45.62 %	42.33 %	44.43 %	47.69 %
	2.53 %	3.04 %	5.09 %	6.71 %	10.38 %	9.86 %	6.24 %	2.67 %	1.17 %	47.69 %
Female	941	1197	2151	2978	5046	4791	3197	1566	629	22496
	4.18 %	5.32 %	9.56 %	13.24 %	22.43 %	21.3 %	14.21 %	6.96 %	2.8 %	100 %
	46.38 %	47.73 %	49.55 %	50.78 %	53.05 %	53.03 %	54.36 %	57.64 %	55.57 %	52.29 %
	2.19 %	2.78 %	5 %	6.92 %	11.73 %	11.14 %	7.43 %	3.64 %	1.46 %	52.29 %
No Answer	1	2	0	1	0	0	1	1	0	6
	16.67 %	33.33 %	0 %	16.67 %	0 %	0 %	16.67 %	16.67 %	0 %	100 %
	0.05 %	0.08 %	0 %	0.02 %	0 %	0 %	0.02 %	0.04 %	0 %	0.01 %
	0 %	0 %	0 %	0 %	0 %	0 %	0 %	0 %	0 %	0 %
Total	2029	2508	4341	5865	9511	9034	5881	2717	1132	43018
	4.72 %	5.83 %	10.09 %	13.63 %	22.11 %	21 %	13.67 %	6.32 %	2.63 %	100 %
	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %	100 %
	4.72 %	5.83 %	10.09 %	13.63 %	22.11 %	21 %	13.67 %	6.32 %	2.63 %	100 %

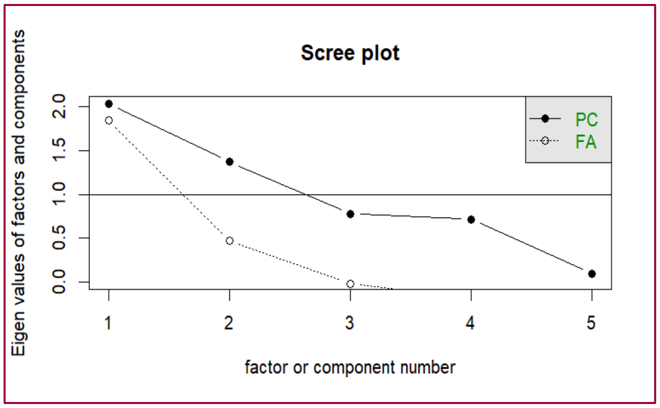
χ^2=131.842 · df=16 · Cramer's V=0.039 · Fisher's p=0.000

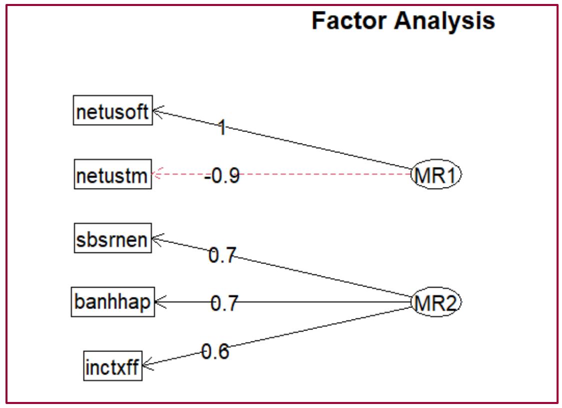
observed values
% within gndr
% within level_concern
% of total

Hypothesis 2: *Individuals who frequently engage in online discussions are more likely to support policies that emphasize environmental sustainability.*

To test the hypothesis that individuals who frequently engage in online discussions are more likely to support policies that emphasize environmental sustainability, I first conducted a **factor analysis** using variables related to online engagement and support for environmental policies. Before proceeding with the factor analysis, I assessed the suitability of the data using the Kaiser-Meyer-Olkin (KMO) measure. The overall KMO value was 0.58, which indicates moderate sampling adequacy. Although this value suggests that the factor analysis may not be ideal, it is still feasible to perform the analysis.

I then performed a scree test to determine the appropriate number of factors to retain. The scree plot indicated a significant drop in eigenvalues after the first factor, suggesting that a single factor might be sufficient. However, I decided to retain two factors due to theoretical considerations related to the hypothesis, which requires an examination of the relationship between online engagement and support for environmental policies.



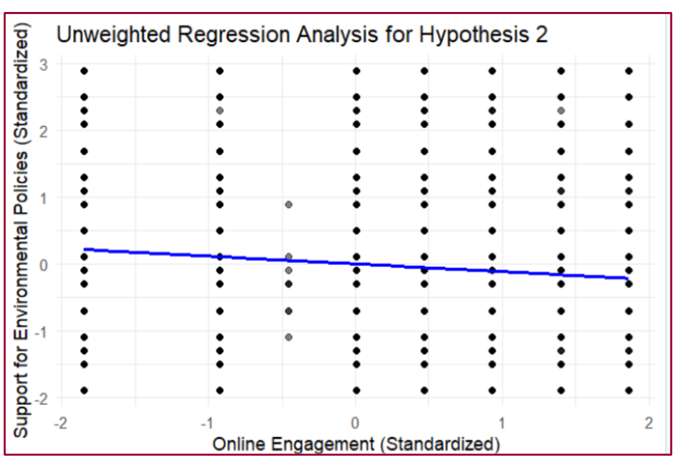


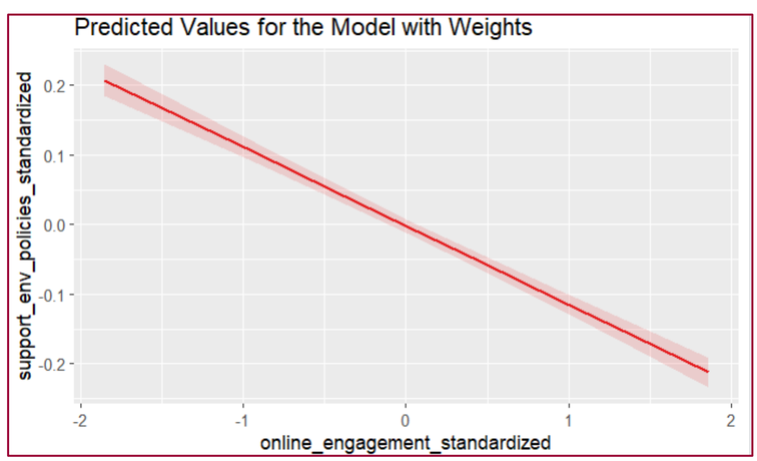
Factor Analysis

	Factor 1	Factor 2
netusoft	1.00	0.00
netustm	-0.89	0.01
inctxff	-0.05	0.56
sbsrnen	-0.00	0.71
banhhap	0.03	0.67

I conducted the factor analysis using an Oblimin rotation to allow for the possibility of correlated factors. The factor loadings were printed with a cutoff of 0.3 for clarity. Factor MR1 seems to represent "Internet Usage." The strong positive loading of “netusoft” and the strong negative loading of “netustm” suggest that individuals who frequently use the internet do not necessarily spend a large amount of time online on a typical day, or vice versa. Factor MR2 appears to represent "Support for Environmental Policies." All three variables related to environmental policies have strong positive loadings, indicating that individuals who support one of these policies are likely to support the others as well. So, in the current results: MR1 (Internet Usage) does not show any loadings for inctxff, sbsrnen, or banhhap. MR2 (Support for Environmental Policies) does not show any loadings for netusoft or netustm. This lack of cross-loading suggests that the factors are more distinct than expected. To investigate further, I have conducted a regression analysis.

I analyzed the relationship between online engagement and support for environmental policies using both **weighted and unweighted bivariate regression** models. In the unweighted model, the slight downward slope indicates a weak negative correlation, but the high variability suggests that online engagement is not a strong predictor. The weighted model also shows a negative relationship, with the regression line indicating that as online engagement increases, support for environmental policies decreases. The narrow confidence interval suggests the relationship is estimated with high precision.



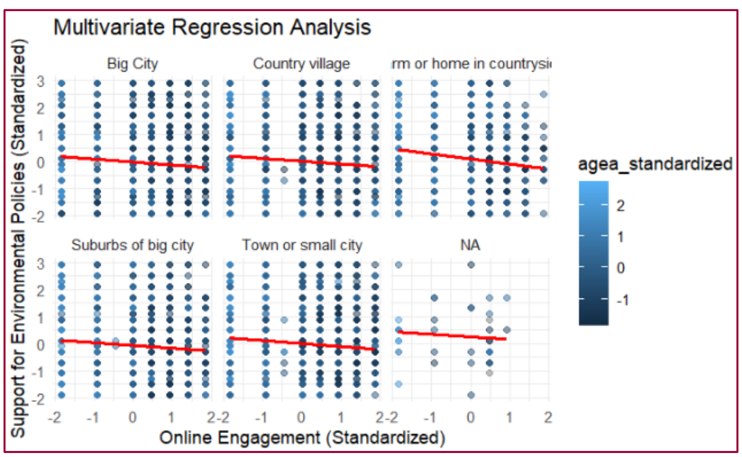


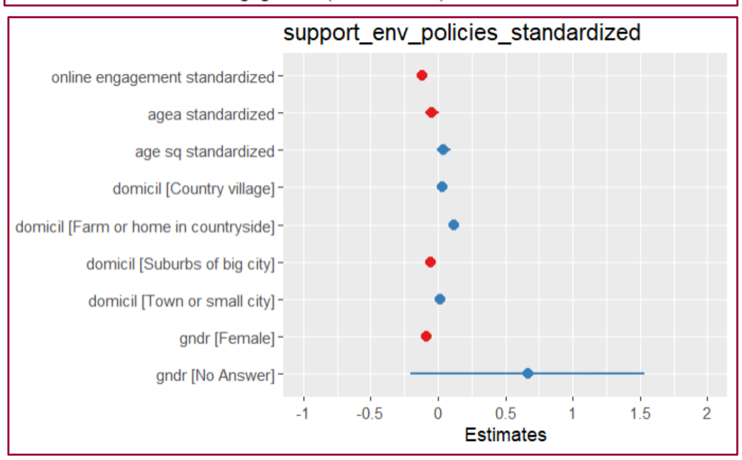
The table below shows that coefficients for online engagement are nearly identical in both models (-0.12 unweighted, -0.11 weighted), indicating a consistent negative relationship with support for environmental policies, regardless of whether survey weights are applied. The small R-squared values (~0.013 unweighted, ~0.012 weighted) suggest that while this relationship is statistically significant, online engagement is not a strong predictor of support for environmental policies.

Table: Comparison of Unweighted and Weighted Bivariate Regression Models for Online Engagement and Support for Environmental Policies.

	Dependent variable	**Dependent variable**
Predictors	*Estimates*	*Estimates*
(Intercept)	0.00	-0.00
online engagement standardized	-0.12 ***	-0.11 ***
Observations	43481	43481
R^2 / R^2 adjusted	0.013 / 0.013	0.012 / 0.012
		* $p<0.05$ ** $p<0.01$ *** $p<0.001$

To gain a more nuanced understanding, I developed a **multivariate regression** model that controls for age, place of residence, and gender. **Age:** The standardized age variable had a non-significant effect, while the squared age variable was significant and positive, indicating a non-linear relationship where support for environmental policies increases with age, particularly in older individuals. **Place of Residence (Domicile):** Compared to those living in big cities, individuals residing in country villages or on farms showed higher support for environmental policies, while those in suburban areas showed lower support. **Gender:** Gender was a significant predictor, with females being less likely to support environmental policies compared to males.





Regression Analysis of Online Engagement and Support for Environmental Policies: Controlling for Age, Residence, and Gender

	support_env_policies_standardized	
Predictors	*Estimates*	*std. Error*
(Intercept)	0.03 *	0.01
online engagement standardized	-0.12 ***	0.01
agea standardized	-0.04	0.03
age sq standardized	0.04	0.03
domicil [Country village]	0.03 *	0.01
domicil [Farm or home in countryside]	0.12 ***	0.02
domicil [Suburbs of big city]	-0.05 **	0.02
domicil [Town or small city]	0.01	0.01
gndr [Female]	-0.08 ***	0.01
gndr [No Answer]	0.67	0.44
Observations	43303	
R^2 / R^2 adjusted	0.016 / 0.016	
		* $p<0.05$ ** $p<0.01$ *** $p<0.001$

Results Interpretation - Part 2: Anusha Maharjan - 6611077

Hypothesis 3*:* *Social Democratic Regimes are more effective at reducing poverty through socio-environmental policies than Liberal and Conservative welfare states.*

Descriptive Analysis

Table 1 describes that Social-Democratic welfare state has the highest average household income with mean recorded as 5.74 and median as high as 6 which suggests that there is less poverty. On the other hand, the most variable income is found in the Conservative welfare state, suggesting that greater income inequality exists in such regimes. When socio-environmental policy variables were added to the income level variables I found that Social-Democratic welfare state demonstrated less support for these environmental policies and less variety while Liberal welfare state demonstrated the most support for instance, raising taxes on fossil fuels and providing subsidies for renewable energy.

Table 1: Descriptive statistics of poverty measures across welfare regimes

Welfare states	Mean Income	Median Income	SD Income	IQR Income
Social-Democratic	5.74	6	2.76	4
Conservative	5.39	5	2.65	5
Liberal	4.38	4	2.58	4
Other	4.97	5	2.75	4

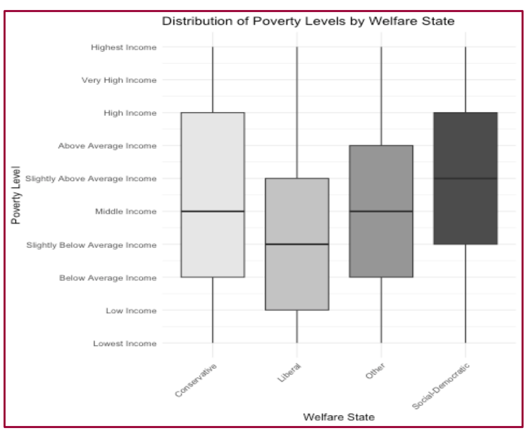


Figure 1: Distribution of poverty levels by welfare state

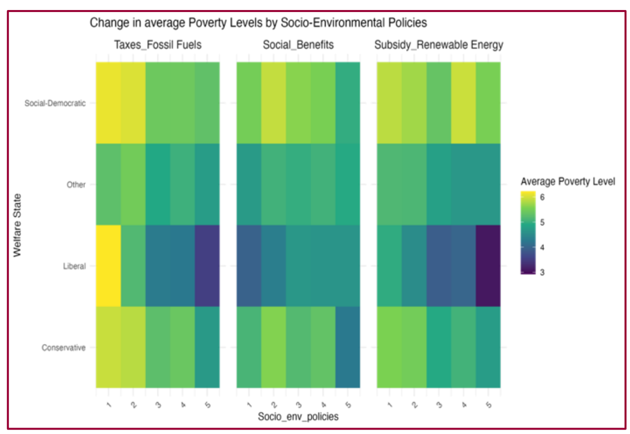


Figure 2: Change in average poverty levels by socio-environmental policies

Figure 1 shows that the income distribution of conservative states, have a significant population concentrated around middle-class income levels. In contrast, there are fewer extreme cases of wealth or poverty in the liberal welfare states, and the concentration of income is more in line with the lower middle range. Interestingly, despite being successful in eliminating extreme poverty, a greater proportion of individuals reside in lower-income groups, challenging the fundamental assumption that the Social-Democratic welfare state significantly reduces poverty and economic disparity. Again, the heatmap in Figure 2 shows that Social-Democratic Welfare State displays moderate average poverty levels across all policies, suggesting that while the policies are implemented, they may not be as effective in significantly reducing poverty. The Liberal Welfare State and Conservative states on the other hand, show lower average levels of poverty, especially when it comes to benefit policies, indicating that socio-environmental policies may be more successful in poverty reduction. Thus, these findings show that my Hypothesis 3 is refuted.

Hypothesis 4: *Individuals with higher socio-economic status (SES) are more likely to use renewable sources of energy compared to those with lower SES.*

To test hypothesis 4, I first conducted a principal factor analysis where I used variable related to SES and renewable energy usage. I assessed the suitability of the data using the Kaiser-Meyer-Olkin (KMO) measure. With KMO scores as low as 0.5, the factor analysis results in Table 3 demonstrate low factor loadings across the variables, suggesting that the data is not appropriate for identifying different underlying factors. I then perform SCREE test to determine number of factors to retain. It is challenging to create useful aggregated indicators because the low factor loadings imply weak correlations between the variables and those identified factors. Due to the poor fit of the factor analysis I move to do regression analysis which allows for a direct examination of the correlations between the socio-economic variables and the use of renewable energy sources without assuming the existence of underlying factors.

Table 2: Factor Analysis

	Factor 1	Factor 2
Education	0.05	0.53
Income Level	0.06	0.65
Paid work	-0.34	0.53
Unemployed	-0.22	-0.33
Retired	1.01	-0.00
Energy use	-0.22	-0.33

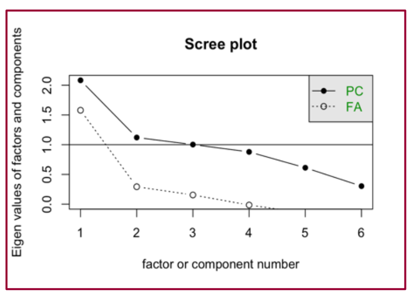


Figure 3: Scree test

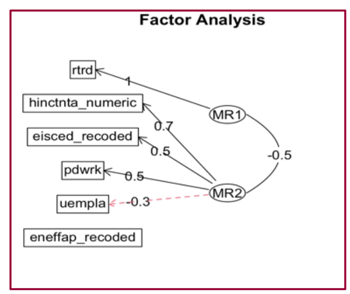


Figure 4: Factor analysis

Regression Analysis

For allowed for bi-variate and multi-variate regression analysis to find the co-relation. I fixed the dependent variable in the study as (usage of renewable energy sources) and independent variables as (highest education attainment), (household income level), (paid work – last 7 days), (retired – last 7 days), (unemployed and actively looking for job- last 7 days). The estimated models are presented in table 3 below:

Table 3: Regression analysis for the use of renewable energy sources by SES

	Model 1	Model 2	Model 3	Model 4
Education	0.060 (0.006)	0.046 ((0.006)	0.047 (0.006)	0.045 (0.006)
Income	0.027 (0.004)	0.028 (0.004)	0.030 (0.005)	
Paid work			- 0.015 ((0.024)	0.238 (0.032)
Unemployed				0.122 (0.062)
Retired				0.431 (0.034)
No. of Obs.	35,073	35, 073	35,073	35,073
R^2	0.003	0.004	0.004	0.009

From Model 1, I know that education has a statistically significant and positive effect, indicating that having more education is linked to a higher probability of utilizing energy-efficient equipment. The addition of household income in Model 2 also demonstrates a positive and substantial effect, albeit a lesser one than that of education. Model 3 includes paid work status, which is not significant, suggesting that, after adjusting for income and education, employment has no discernible effect on the use of energy-efficient appliances. Lastly, retirement status and unemployment are included in Model 4, while unemployment has a tiny positive effect, retirement has a big positive effect, suggesting that retirees are substantially more inclined to use energy-efficient equipment. All models have low R-squared values, suggesting that although these factors have statistically significant effects, they explain only a small portion of the variation in energy-efficient appliance usage.

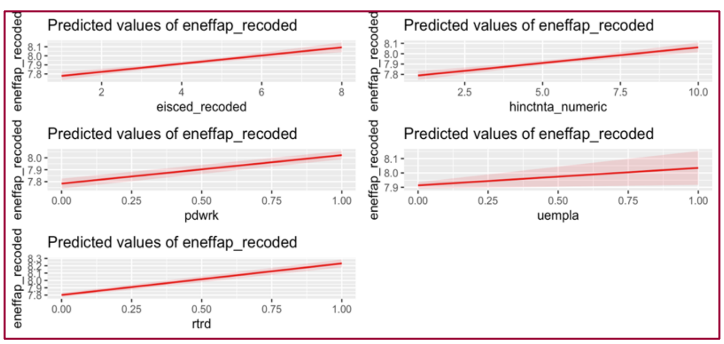


Figure 6: Plot prediction of Model (1-4), effect of SES on of renewable sources of energy

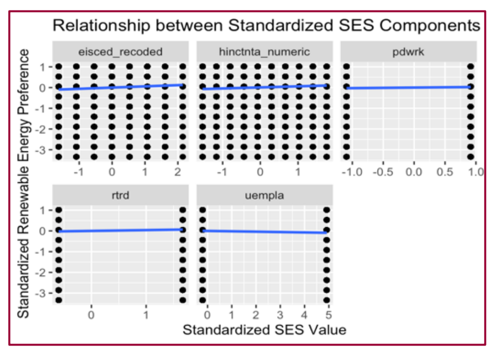


Figure 5: Standardized SES values use

In Figure 6, the prediction plot shows that higher education, income, paid work, and retirement are associated with increased use of energy-efficient appliances, as indicated by the upward-trending lines. The effect of unemployment is less pronounced, with a flatter line. The shaded areas around the lines represent the confidence intervals, highlighting the uncertainty in predictions. The highest use of alternative energy sources can be seen amongst retirees than any other group. Now to check the robustness of the analysis I standardized the values and the visual representation in Figure 5, the blue trend lines are nearly flat, suggesting there is no big significant linear relationship between the standardized SES components and standardized renewable energy preference. Different SES components (education, income, work status, retirement status, and unemployment) have minimal impact on the renewable energy preferences but not that quite strong. The results thus confirm hypothesis 4, which states that people with greater socioeconomic status—which includes factors like income, education, and employment—use energy-efficient applications more frequently than people with lower socioeconomic status, who use less renewable energy sources.